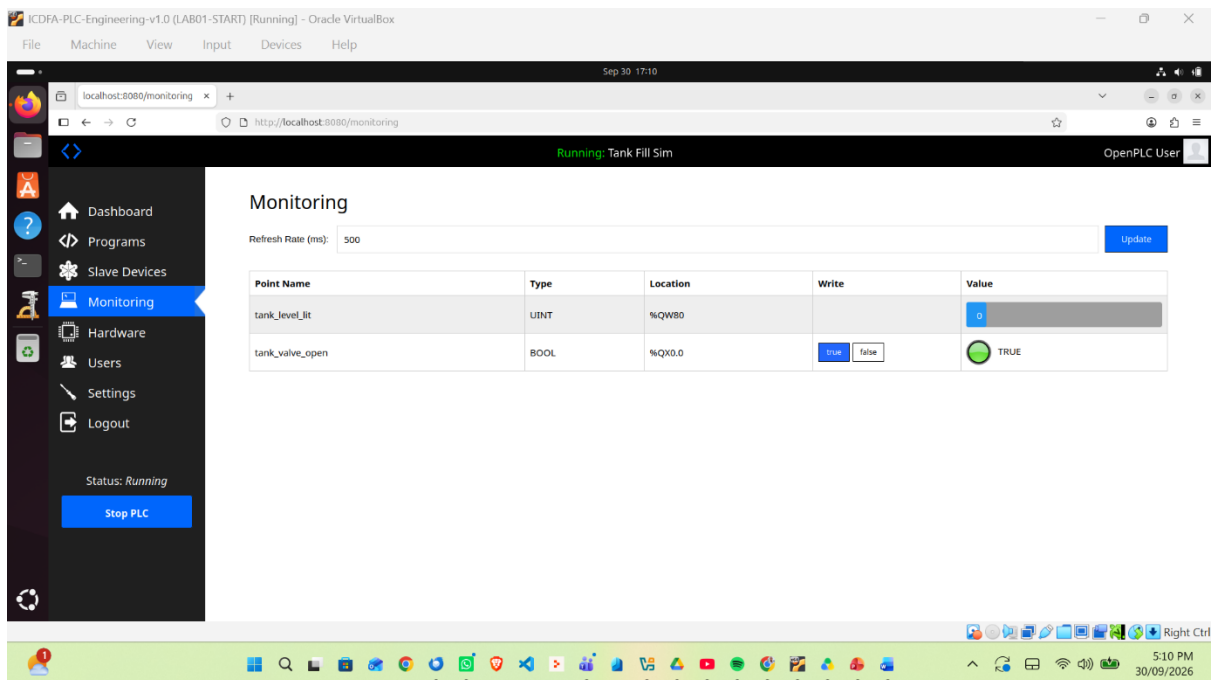


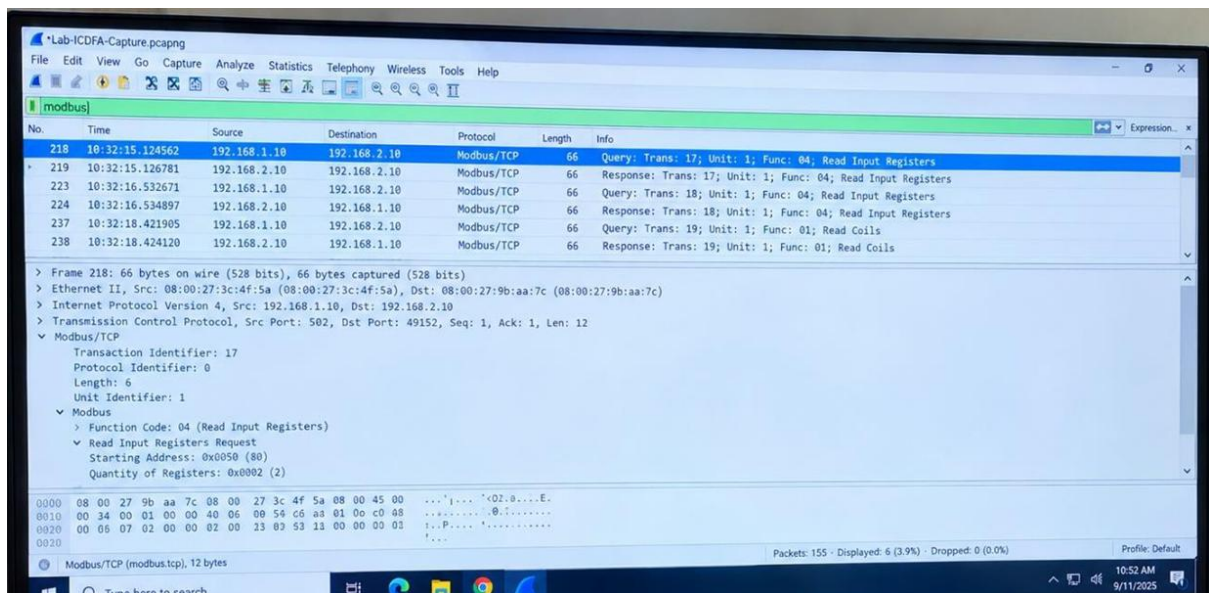
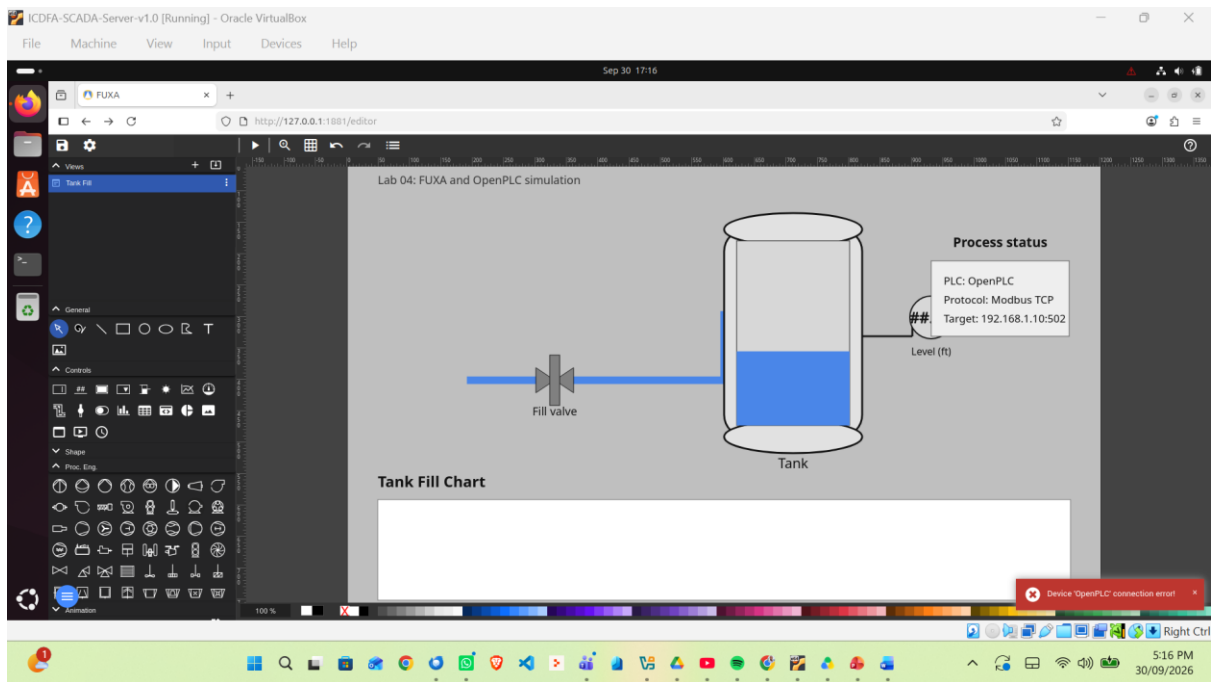
LAB REPORT

Lab 06: Introduction to Modbus

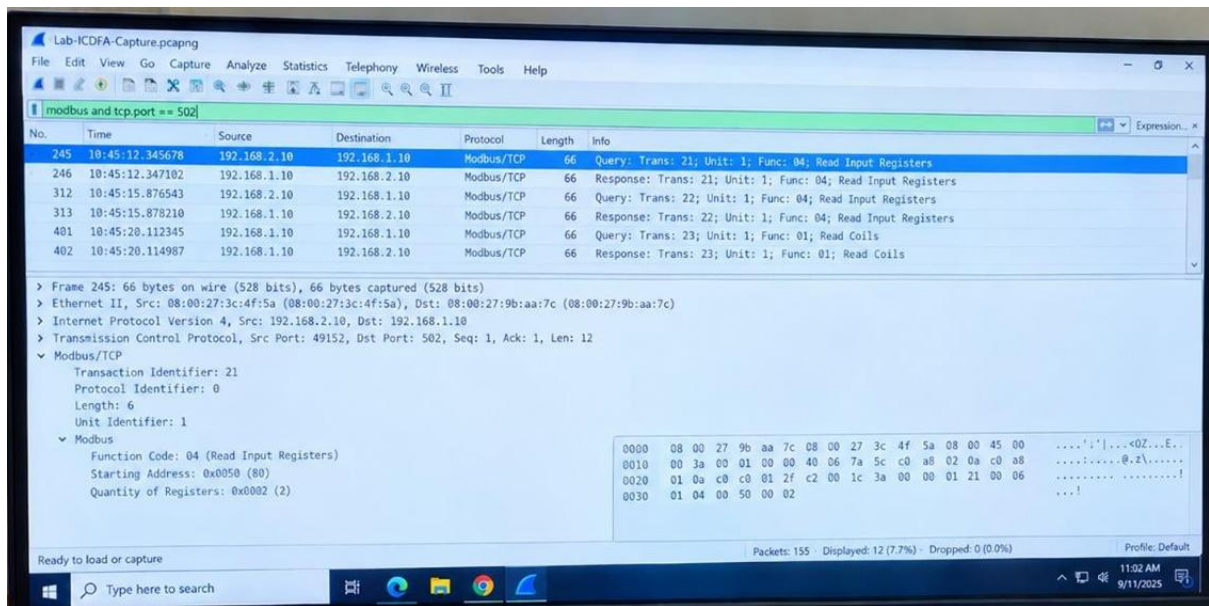
Lab Title	Lab 06: Introduction to Modbus
Student Name	Adamu Abubakar
Student ID (Reg No)	C11/26/ACIS/17353
Date of Submission	01-10-2026



Started the PLC program. On the PLC VM I started TankFillSim in OpenPLC Runtime and confirmed the status is Running.



Capture the traffic. I started a short capture while the HMI was running. I waited for several process updates, then stopped the capture.



Study one request and response. I selected one Modbus request and its response pair. I expanded the **Modbus/TCP** and **Modbus** fields in the Packet Details pane. I identified the transaction, the source and destination IP addresses, the function code and the register and value.

Annotated Request/Response Analysis Example Modbus TCP Request

Field	Value
Protocol	Modbus TCP
Source	SCADA/HMI Client
Destination	PLC Server
Function Code	04
Operation	Read Input Registers
Register	Input Register Address (Example: 0)

Modbus TCP vs Modbus RTU Comparison Table

Feature	Modbus TCP	Modbus RTU
Communication Medium	Ethernet	Serial communication
Network Type	TCP/IP network	Serial network
Addressing	IP addresses	Device addresses
Architecture	Client-Server	Primary-Secondary
Physical Layer	Ethernet	RS232 / RS485
Common Usage	Modern SCADA systems	Older industrial serial systems

Knowledge Check

1. What transport/networking difference separates Modbus TCP from Modbus RTU?

Modbus TCP uses Ethernet and TCP/IP network communication, where devices communicate using IP addresses. Modbus RTU uses serial communication, typically through RS232 or RS485 connections. Modbus TCP is commonly used in modern industrial networks, while Modbus RTU is used for serial-based industrial devices.

2. Why does Modbus use function codes?

Modbus uses function codes to define the specific action to perform on data. They tell the client what operation is required, such as reading coils, reading registers, or writing values to the PLC memory. The function code appears before the data section in the Modbus frame.

3. What does Function Code 4 request/read?

Function Code 04 requests a read operation of Input Registers. It reads 16-bit input register values from the Modbus device.

4. Why is visibility of unprotected industrial protocol traffic a cybersecurity concern?

Unprotected Modbus traffic can expose communication between SCADA systems, PLCs, and industrial devices. Attackers who can observe the traffic may identify device addresses, function codes, and data being transmitted, which could allow unauthorized monitoring or manipulation of industrial processes. Modbus defines how data is organized for communication, but it does not provide built-in security protections such as encryption.